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Apps for capturing language variation and change in German-speaking Europe: Opportunities, challenges, findings, and future directions

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Abstract: Smartphone apps are ubiquitous: in 2018 in Germany alone, 79% of the population owned a smartphone. Virtually everyone in this cohort always carries with them a recording device and a notepad (i.e., a screen), two essential tools that linguists typically use in the field. In the present contribution we discuss how linguists can harness this ubiquity of linguistic tools in the population to capture language variation and change, illustrated through apps that were developed for German-speaking Europe. We present four apps (*Dialäkt Äpp*, *Voice Äpp*, *Grüezi Moin Servus*, and *Deutschklang*) that were developed to (a) engage with the public and (b) to collect linguistic data. We discuss opportunities (e.g., the multimodality of said devices), as well as challenges (e.g., maintenance, updating and the costs involved therein). Finally, we present new findings that have emerged from working with this new paradigm and speculate about future directions and developments in using smartphone apps to collect linguistic data.

Keywords: smartphone; methods; phonetics; German; dialects

1 Introduction

Methods in sociolinguistics have been evolving since the beginning of the discipline in the 1960s. Typical fieldwork methods have included surveys, interviews, questionnaires, ethnographic and participant observation, and systematic note-taking and record keeping (Hernández-Campoy 2014). Notepads and microphones have been the central tool of linguistic data collection. Today, the majority of citizens of westernized countries have access to a smartphone, and thereby a microphone and notepad, facilitating the collection of linguistic data: in Germany alone in 2018, 79% of the population owned a smartphone (Newzoo 2019). Global coverage is uneven: in 2018, smartphone user penetration as percentage of total global population was 34.7%, predicted to rise to 40% by 2021 (Statista 2019).

Given the ubiquity of smartphones, we wish to discuss how smartphones can be exploited for linguistic data collection, particularly in studies of language variation and change. In 2012, we began developing a smartphone app for Swiss German, called *Dialäkt Äpp* (Kolly and Leemann 2013; Leemann 2013). *Dialäkt Äpp* has two core functionalities: predicting where the user is from based on their choice of dialect words and recording the user's dialect. Originally, the app was developed as a means of engaging with the public – without us having scientific goals in mind. We soon noticed that the app is well received by the academic community, and served as a springboard for a Swiss National Science Foundation (SNSF) funded science communication project called *Voice Äpp* (Leemann et al. 2015b; Leemann 2015b). Aside from predicting a user's regional origin within German-speaking Switzerland using automatic speech recognition, *Voice Äpp* features an analysis of speaking rate and fundamental frequency, illustrates hearing pathologies, speech phenomena like the Cocktail Party Effect (Cherry 1953) and McGurk Effect (McGurk and MacDonald 1976), and provides an interactive introduction to Swiss German dialectology. Following *Dialäkt Äpp* and *Voice Äpp*, the smartphone and web-app *Grüezi, Moin Servus* (Leemann 2015; Leemann et al. 2018b) was developed in collaboration with Zurich-based *Tagesanzeiger* and *Spiegel Online* to predict users' regional origins within all of

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German-speaking Europe. The smartphone app further features recording capability as well as dialect evaluation experiments. Finally, *Deutschklang* (Leemann 2016a) has provided a reduced functionality for users to record sentences in their regional accent. The same team created the *English Dialects App* in 2016, which is discussed elsewhere in this special collection (see Blaxter and Britain this issue).

This contribution pursues the following goals: first, it will present these four apps and show their core functionalities in the context of linguistic data collection. The following sections will discuss the opportunities and challenges of these apps, i.e., how do smartphone apps provide an enhanced data collection paradigm? What are some of the limitations and pitfalls of smartphone app-based data collection? The section on challenges, in particular, will hone in on lessons learnt from developing these apps for linguistic data collection. We will also introduce some new findings that have emerged from this paradigm before we attempt a look into the future, discussing potential future directions and trends in using apps for linguistic data collection.

2 Overview of apps

2.1 Dialäkt Äpp

Dialäkt Äpp (Kolly and Leemann 2013; Leemann 2013) was the first in a series of free apps developed for iOS to (a) collect linguistic data and (b) engage with a broader public about linguistic research. Development of the app was paid for by a crowdfunding campaign that raised 10,000 CHF (see Video 1 [see Supplementary material] for promotional material). The app has two core functionalities: (i) to predict the user's regional origin, and (ii) to enable users to record their own speech. (i) is based on dialect maps from the *Linguistic Atlas of German-speaking Switzerland* (1962–2003) and was implemented in such a way that the user can indicate their dialectal variant for 16 words (Figure 1, left panel). Users are then shown their predicted regional origins with a map showing five pins (Figure 1, second panel). In the second functionality, users can record dialect words and a short reading passage (Dieth 1952). Each locality with a recording is indicated as a pin on the map (Figure 1, third and rightmost panels). Audio 1 (see Supplementary material) is a recording of the reading passage, spoken by a 47-year-old male from Chur (Canton of Grisons).

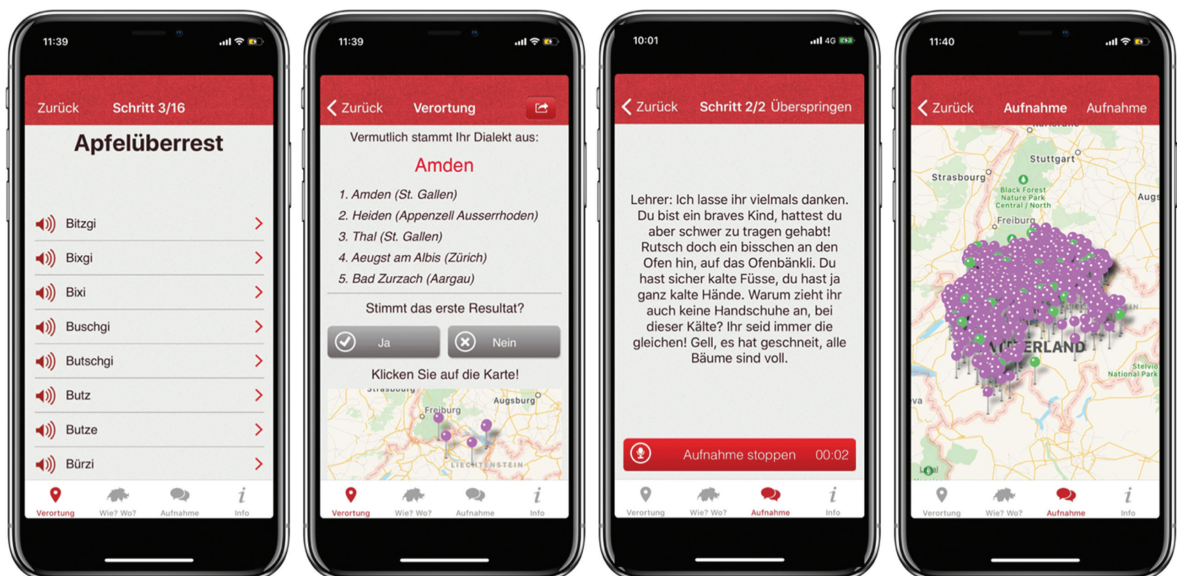


Figure 1: Screenshots for dialect prediction functionality in *Dialäkt Äpp*: selection of variants (left) and result of localization (second from left). Screenshot for recording functionality: reading prompt (second from right), and map that shows places with recordings (right).

When evaluating the quiz results and participating in the recordings, users were asked to provide meta-data on their age, gender, and regional origin (as a drop-down menu using a set of 550 localities). The app was made public with a press release from the University of Zurich, which was picked up by national and international media. Upon its launch, the app was the most-downloaded app in the country (Kolly and Leemann 2013). Since its inception, it has been downloaded 104,549 times, 92,503 have participated in the quiz, and 4259 speakers have provided audio recordings (as of 25 February 2019).

2.2 Voice Äpp

Voice Äpp (Leemann et al. 2015b; Leemann 2015b) was a free app specifically designed for iOS and Android devices to interact with the public and serve as a means of science communication. The app was funded with a grant from the Swiss National Science Foundation (“Agora” scheme). The app has two core functionalities: (i) “My Dialect” and (ii) “My Voice”. The first functionality predicts the user’s regional origin. It does so through automatic speech recognition: users articulate a set of 15 words (e.g., “apple core”) (Figure 2, leftmost panel) and the app shows a heatmap with the most probable geographic origin (Figure 2, second panel). The speech recognizer was built with HTK (Young et al. 2006), using word recordings from *Dialäkt Äpp* (see Section 2.1) as training material. The second core functionality enables users to record a sentence, based on which the app analyzes articulation rate (Figure 2, third panel) and pitch (Figure 2, rightmost panel) and compares these to the other participants. Further functionalities include an illustration of hearing pathologies, e.g., having tinnitus, a general introduction to regional variation in Swiss German dialects, an introduction to the Cocktail Party (Cherry 1953) and McGurk Effects (McGurk and MacDonald 1976), as well as a “dialect panorama” – a functionality that uses augmented reality to illustrate regional variation (see Section 6). Audio 2 (see Supplementary material) is a recording of a user to which the app added a tinnitus signal (in this case a high-pitched sine wave) of 8 kHz and damped higher frequencies to simulate mild hearing loss.

Here, too, participants indicated metadata on their regional origins (by postal code), age, and gender. The app was advertised with a press release from the University of Zurich and a promotional video (see Video 2 [see Supplementary material]). Since its inception, 23,880 users (iOS 18,106; Android 5774) have downloaded the app (as of 25 February 2019).



Figure 2: Screenshots of “My Dialect” functionality in *Voice Äpp*: recording of variants (left) and result of localization (second from left). Screenshots of “My Voice” functionality: analysis of speaking rate (second from right) and pitch (right).

2.3 Grüezi, Moin, Servus

In early 2014, we were approached by a data journalist at the Zurich-based *Tagesanzeiger*. The journalist had heard of the success of *Dialäkt Äpp* and the NYT's dialect quiz, a dialect prediction quiz for American English (NYT 2013). He wanted to run a similar story with his newspaper. We told him that we were working on a dialect quiz that predicts users' regional origins within all of German-speaking Europe. After a number of meetings, we decided to develop a web-app dialect quiz in collaboration with *Tagesanzeiger* and *Spiegel Online* (Leemann et al. 2015). At the same time, we developed a 99¢ smartphone app for iOS and Android that – aside from the same dialect quiz as the web-app – also included recording capability and dialect evaluation experiments (Leemann et al. 2018a). Web-app development expenses were covered by the newspapers, while development of the accompanying smartphone app was paid for with personal money. To work out a prediction for all of German-speaking Europe, we used maps from the *Atlas zur Alltagssprache* (Möller and Elspaß 2015). Twenty-four variables, such as the choice of grammatical gender and lexeme for “paprika” (Figure 3, leftmost panel), were selected. The user is shown a heatmap that shows the most probable geographical origin within German-speaking Europe (Figure 3, second panel). The web-app features only this functionality (Figure 4).

The smartphone app further features a recording functionality that prompts users to record 12 sentences taken from Wenker's Atlas for Germany (Schmidt and Herrgen 2005–2009). Users can then listen to other users' recordings (Figure 3, second panel). Audio 3 (see Supplementary material) is a recording of the prompt shown in Figure 3 (second panel from the left), spoken by a male speaker from Berlin. Finally, we included a new functionality: perception experiments. In one experiment, users are asked to evaluate tokens (based on *Dialäkt Äpp* recordings) with different types of /l/ qualities (e.g., lateral, velarized, vocalized) for pleasantness judgments by adjusting a lever on a scale from “very pleasant” to “very unpleasant”. In the other experiment, users evaluate the pleasantness of different accents, with stimuli from Leemann et al. (2015) (Figure 3, rightmost panel).

For both the web and smartphone apps, users provide metadata on age and gender; users also place a pin on a map to provide their regional origin. The web-app was launched on 25 April 2015. Within only a few days, it attracted more than a million users (Brupbacher 2015). The smartphone app performed less favorably – predictably so because (a) the app was not free and (b) the core functionality – prediction of regional origin – was already included in the free web-app. Since its inception, 8078 users (iOS 5690; Android

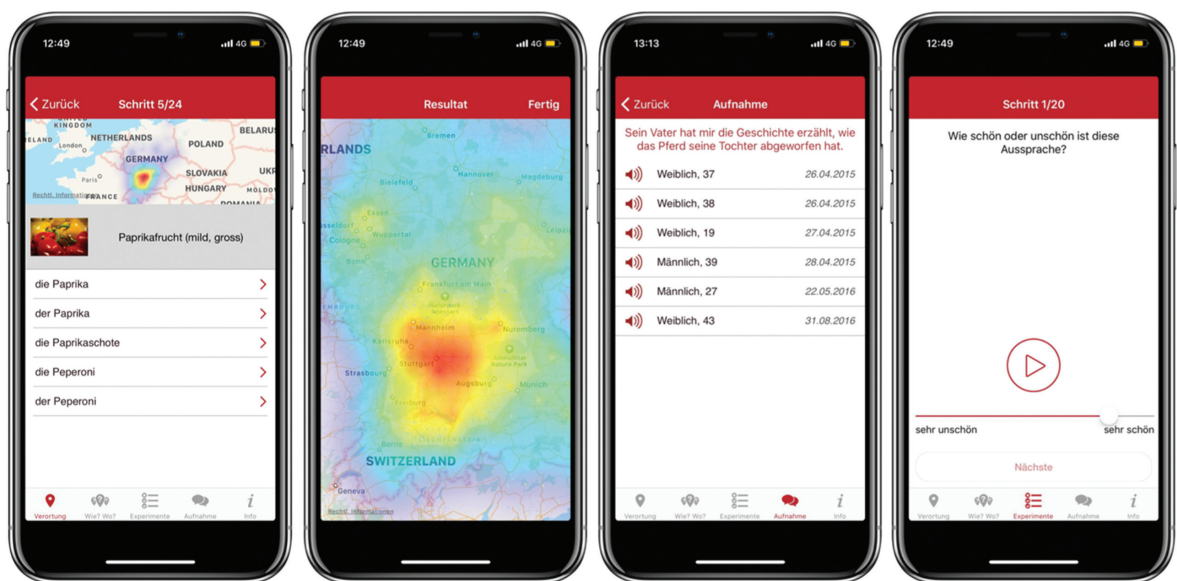


Figure 3: Screenshots for dialect prediction in *Grüezi Moin Servus* – selection of variants (left), result of localization (second from left), playback of audio recordings (second from right), and dialect evaluation experiment (right).

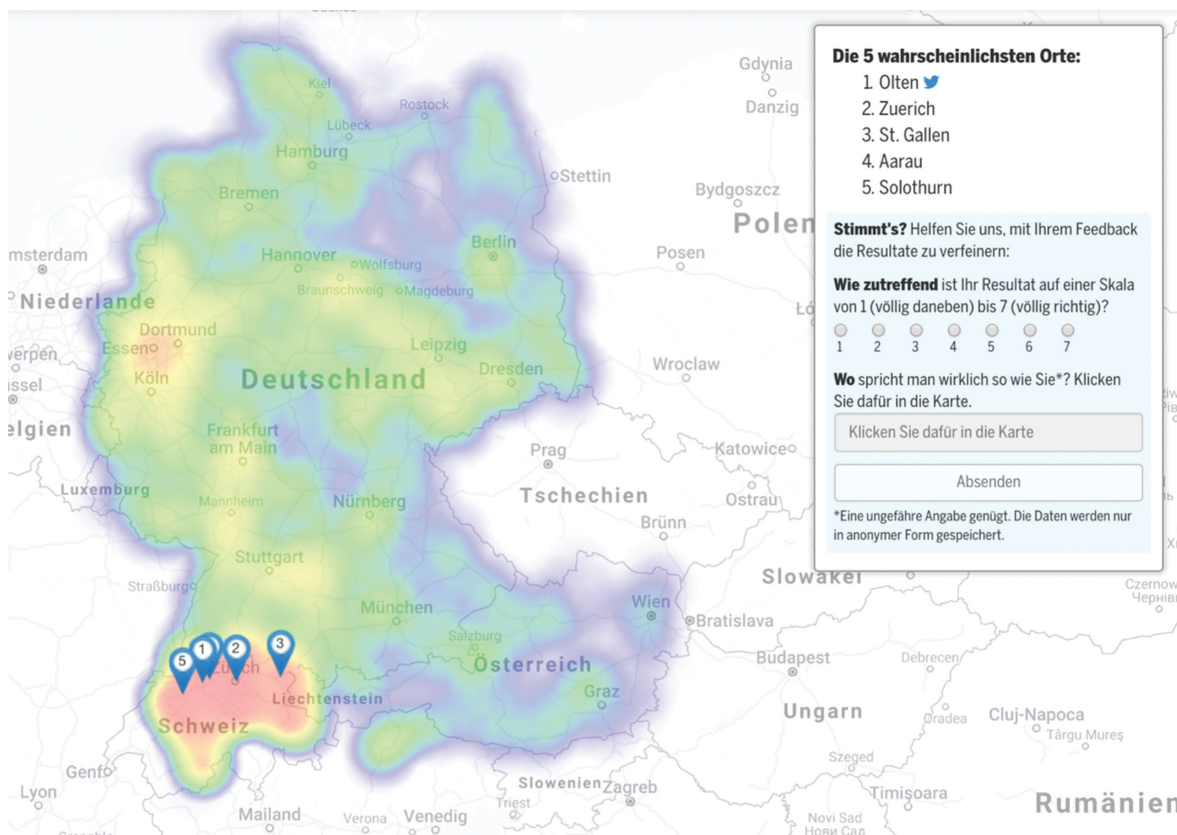


Figure 4: Screenshot for dialect prediction functionality in *Gräezi Moin Servus* web-app.

2388) have downloaded the smartphone app. More than 2 million users have participated in the web-app quiz.

2.4 Deutschklang

After the success of the *Gräezi Moin Servus* web-app and the less-popular smartphone app (which had recording capability), we tried to rekindle public interest with a new app, called *Deutschklang* (Leemann 2016a). *Deutschklang* was developed for free download on iOS and Android, with audio recording capability only. Development of the app was paid for privately. First, users indicate metadata before proceeding to the recording instructions (Figure 5, second panel). This time, however, the metadata information requested is more elaborate – aside from age, gender and region, we ask for information on speaker mobility and education background, which was inspired by the *English Dialects* app (Leemann et al. 2016b). Users are then prompted to record 12 sentences (Figure 5, third panel). The same Wenker sentences as in the *Gräezi Moin Servus* app are given to the users as prompts. Audio 4 (see Supplementary material) is a recording of the sentence shown in Figure 5 (second panel from the right), spoken by a male from Graz.

In all apps, users have the option to view the privacy statement (e.g., Figure 5, right panel). The app was made public by embedding information about it in an online article that appeared in *Spiegel Online* (Elmer et al. 2016) and *Schweizer Radio und Fernsehen* (Grossenbacher and Stotz 2016). The article showed maps illustrating regional variation in German-speaking Europe which were based on the data collected through the *Gräezi Moin Servus* web-app in the previous year (Figure 6).

Mentioning of the app spiked some interest and more than a thousand users have provided sentence recordings (total number of downloads 4084: iOS 2000; Android 2084; 1428 speakers have provided sentence recordings).

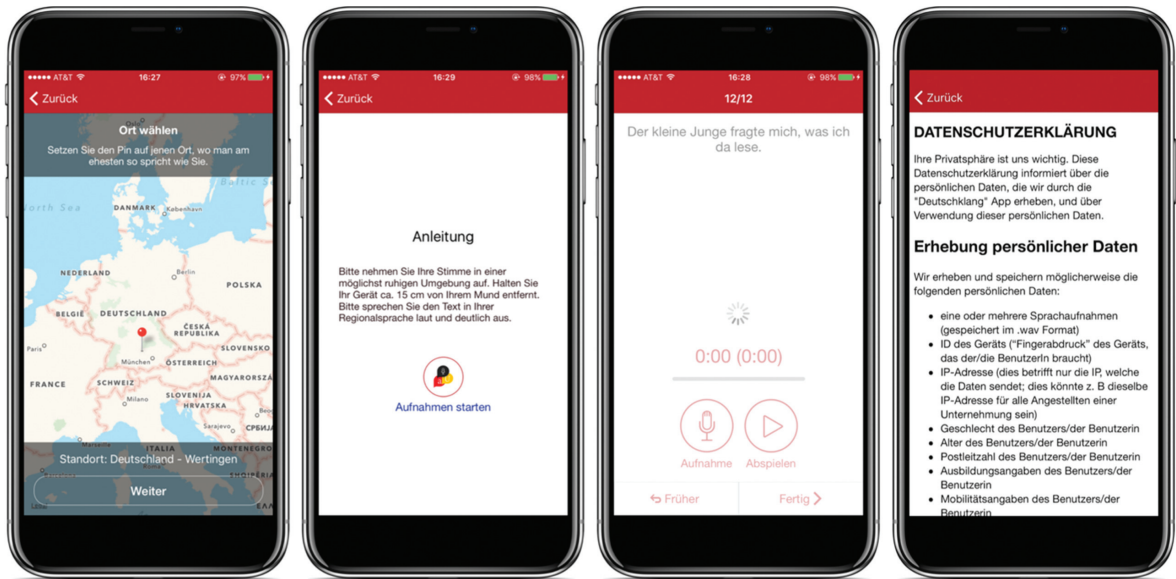


Figure 5: Screenshots for dialect recordings in *Deutschklang*: elicitation of user's regional origin (left), instructions for recordings (second from left), recording prompt (second from right), and privacy statement (right).

SPIEGEL ONLINE SPIEGEL



Anmelden

Alltagssprache

Hier verlassen Sie den Bulettensektor

Ihr Vokabular verrät, woher Sie stammen. Unsere interaktive Karte zeigt, wo die Grenzen liegen - zwischen Pfann- und Eierkuchen, Bolzen und Kicken, Frikadelle und Fleischpflanzerl.

Von *Christina Elmer* ▼, *Timo Grossenbacher* und *Patrick Stotz* ▼

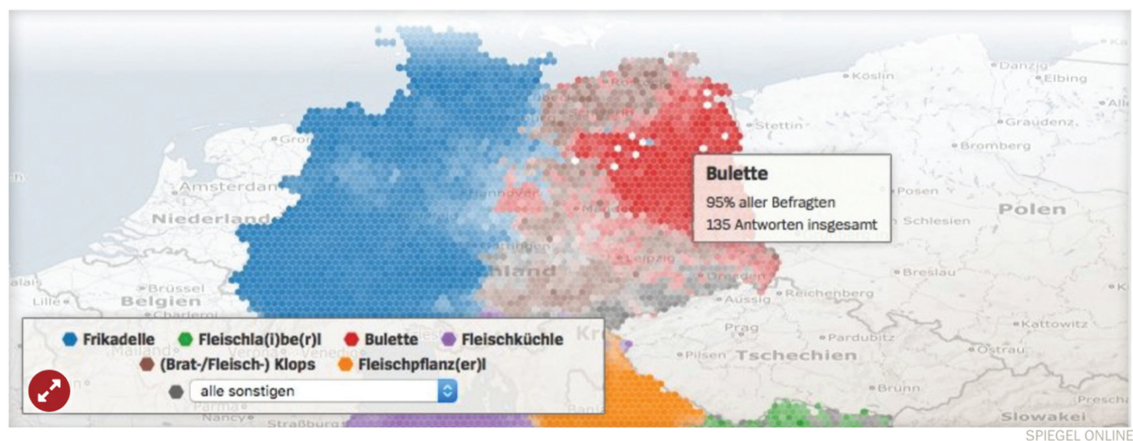


Figure 6: Map of “beef patty” variants for German-speaking Europe. Published by Spiegel Online (2016).

3 Opportunities

Using smartphones for linguistic data collection brings numerous opportunities. Smartphones are ubiquitous and multimodal (including screens, speakers, mics, cameras, etc.). Ubiquity and multimodality are probably

tightly connected – the more functionalities a smartphone has, the more likely the device is being used. Consequently, most individuals in the Western world always carry with them a microphone, a keyboard, a camera, a speaker, and a screen. It is exactly some of these tools that researchers bring with them when going into the field to collect linguistic data. These features make smartphones an ideal tool for data collection not only in linguistics, but also in medicine (cf. Buchholz et al. 2016), geography (Birenboim and Shoval 2016), psychology (cf. Harari et al. 2016), and education research (Anshari et al. 2017). Because of the fact that smartphones are so omnipresent, apps that are devised for research can – if disseminated properly – have wide reach in the population. In return, this can boost societal impact. University activity assessments like the TEF (Teaching Excellence Framework), REF (Research Excellence Framework), and its latest addition KEF (Knowledge Exchange Framework) actively promote such wide-impact research programs (cf. Britain et al. 2018). KEF, for example, evaluates how researchers engage with and transfer knowledge to the broader public. The apps introduced here engage with the public through linguistic games and experiments, but also by providing clear information (e.g., about dialects and hearing pathologies). From a scientific point of view, the unprecedented scale of data collection facilitated by smartphone apps is a great benefit. Having an “open call” for participation further has the often-positive consequence of taking the judgment and sampling of “authentic speakers” out of the hands of potentially biased researchers (Britain 2015; Leemann et al. 2018).

4 Challenges

We also wish to address some of the more challenging aspects of collecting linguistic data with smartphone apps. The problems to be discussed are threefold: (i) How are users recruited? (ii) What are some of the more general pitfalls? (iii) What issues with app development and maintenance appear when using smartphones as a methodological tool?

As for (i), attracting users to participate in an app is a difficult task. The keywords here, from our point of view, are “attention” and “reward”. “Attention” is probably the most valuable resource in today’s app market. Thankfully, humans tend to be interested in the way they speak. Regional variation, in particular, is often a topic when people first meet. Further, users typically seek some form of reward – ideally while partaking in a game – whether in the form of reaching a new level in Candy Crush or learning something about their linguistic origins. Looking at the user statistics (see Section 2), it is clear that the apps that included dialect prediction in a gamified way fared best. When the user is just asked to do a task without it being part of a game, the app is not likely to fare well (e.g., *Deutschklang*, *Voice Äpp*). Gamification – which Deterding et al. (2016) define as “the use of game design elements in non-game contexts” appears to be key for a linguistic app that wants to maximize its reach. It helps, of course, if the app is publicized by university press offices and media partners.

As for (ii), myriad general pitfalls are worth mentioning. Writing instructions, for example, is one of the trickiest parts of the paradigm. While researchers in the field can interact with participants and clarify questions, smartphone users only have in front of them what is shown on the screen. What is more, users typically leave sites within short periods of time (e.g., websites: 10–20 s, Nielsen 2011). Further, the audio recordings collected can have background noise – i.e., are recorded in uncontrolled conditions – which can impinge on acoustic analyses. In *Voice Äpp*, for example, 5–10% of the recordings featured background noise and/or had participants playing around (Leemann et al. 2015b). Participants may further be distracted by incoming messages. This heavily contrasts with experiments taking place in sound-attenuated booths (Miller 2012). When listening to pre-canned recordings for the selection of dialect variants, for example, users are essentially performing a speech perception task. Listening to stimuli with background noise present – i.e. in uncontrolled, non-laboratory conditions – is, for obvious reasons, problematic. There are also issues of multiple submissions – users may participate several times on their own device or on someone else’s device (Birnbbaum 2004). Reips (2002) suggests, however, that for internet-based research, this number is typically <3%. There are further questions regarding the validity of people’s intuitions about their own pronunciations. How well can we trust speakers to judge which regional variant they think they use (cf. Labov 1996)? There may also be

non-native speakers taking part in the app, or speakers faking regional accents. Finally, there is the issue of obtaining biased datasets when making the app public through university press releases: the target audience is typically young and well-educated (cf. Leemann et al. 2018). All of these factors contribute to noise accumulation in the data and call for some form of data validation and quality control. As a start toward data validation, Leemann et al. (2016) have shown that using smartphones to study the diffusion of /l/-vocalization in Swiss German, for example, yields very similar results to studying this phenomenon with more traditional sociolinguistic techniques (e.g., rapid anonymous surveys).

As for (iii), app development and maintenance involve a number of issues. The greatest costs involved once the apps have been published is regular maintenance in order for them to work on Android and iOS, as Apple and Google regularly update their operating systems. Apple, for example, updates iOS almost on a yearly basis, with incremental updates to fix minor bugs in between. Chances are that the apps that were developed for one OS no longer work for subsequent OSs. This raises the question: when apps are funded with government grants (like *Voice Äpp*), for example, how long should these apps be running for? How much of the costing should be allocated to cover such updates? *Voice Äpp* was withdrawn from iOS and Google Play because updates were no longer viable. Data storage, too, is a long-term investment. Collectively, these apps use c. 150 GB of server space, which needs to be paid for on a yearly basis. In terms of privacy, at this stage users can view the respective privacy statement in the app, where it is outlined that the data provided can be used for scientific purposes. This type of data collection and subsequent analyses conform to Swiss federal and cantonal laws. We did not seek further ethical approval from other respective bodies. With the introduction of General Data Protection Regulation (GDPR) in Europe, perhaps these privacy statements need to be amended in the future.

5 New findings

These apps have generated linguistic data on an unprecedented scale. Accordingly, there are countless possibilities for using the data for explorative and research purposes. In what follows we do not intend to summarize the studies that have already been published using the data; instead, we showcase some of the new findings that have emerged from student projects, popular science work, and work which has been published only as abstracts at conferences. We particularly focus on data collected through *Dialäkt Äpp*, *Grüezi Moin Servus*, and *Deutschklang*.

Data crowdsourced through *Dialäkt Äpp* have provided fertile ground for a number of student projects that have not previously been published. In one such study, Renning 2013a looked at the influence of background music on the perception of speaker age. Renning selected read text recordings from six speakers from *Dialäkt Äpp* (recording functionality) and presented these to 91 listeners in three different conditions: in one condition there was no background music, in condition two she added pop music, and in condition three she added classical music. She found that listeners perceived speakers as younger (by an average of 1–3.5 years) if pop music was added. Renning's study contributes to the body of literature that shows how listeners use contextual information, such as background noise, when assessing a speaker's age. In another study, Renning (2013b) examined whether the perception of speaker age is contingent on the regional origin of listeners. She recruited 121 participants (65 Swiss Germans, 56 Germans) and played them audio stimuli retrieved from 30 *Dialäkt Äpp* speakers (audio functionality) of different age groups. Renning did not find an effect of listener origin, concluding that a listener's familiarity with a speaker does not substantially influence their judgment of the speaker's age. Hasler (2013) used the dialect quiz corpus to compare more than 7000 Bern Swiss German users' variants to those found in the historic *Linguistic Atlas of German-speaking Switzerland* (1962–2003). For some variables, Hasler found substantial change (e.g., dialect words for "apple core"); while other variables remained more stable (e.g., /l/-vocalization). Hasler contextualizes the change within Trudgill's gravity framework (Chambers and Trudgill 1980). Most recently, Kolly et al. (2019) have made use of *Dialäkt Äpp*'s quiz data to visualize apparent time change for a handful of Swiss German localities in the context of a story for the *Neue Zürcher Zeitung*.

The data from *Grüezi Moin Servus*' web-app has been particularly fruitful. Leemann et al. (2018b) used the quiz evaluations to write a popular science book. Based on data from more than 700,000 users coming from more than 18,000 localities, maps and stories about these variables were prepared as a coffee table book. Leemann et al. (2018) report on one specific trend in a number of the maps presented in the book: larger cities typically orient themselves more towards supralocal variants, while adjacent rural regions appear to be using more traditional forms. To name just one example, different variants for the variable “beef patty” are widespread in Northeastern Germany. *Bulette* (green), in particular, but also *Klops* (pink) are dominant, see Figure 7. Larger cities like Leipzig, Magdeburg, and Dresden, which are geographically within *Klops* (and in the case of Dresden *Beefsteak*) regions, however, appear to be using the supralocal *Bulette*, with Berlin as its center of usage.

The finding that innovative features like *Bulette* spread from populous dominant centers to other nearby towns and cities, surpassing rural regions in between, has of course long been known and reported, e.g., in gravity diffusion in Great Britain (cf. Britain 2002). Elspaß et al. 2018 further report evidence for change in apparent time using the corpus at hand. In another project with colleagues at the University of Southern Denmark and the University of Bristol, the web-app corpus is being used to explore changing cultural borders and economic interactions in German-speaking Europe. The project operates on the basis of the assumption that cultural change is a slow process that is reflected in the linguistic changes of a language. The data collected through the experiments and the recording functionality included in the smartphone app have yet to be analyzed.

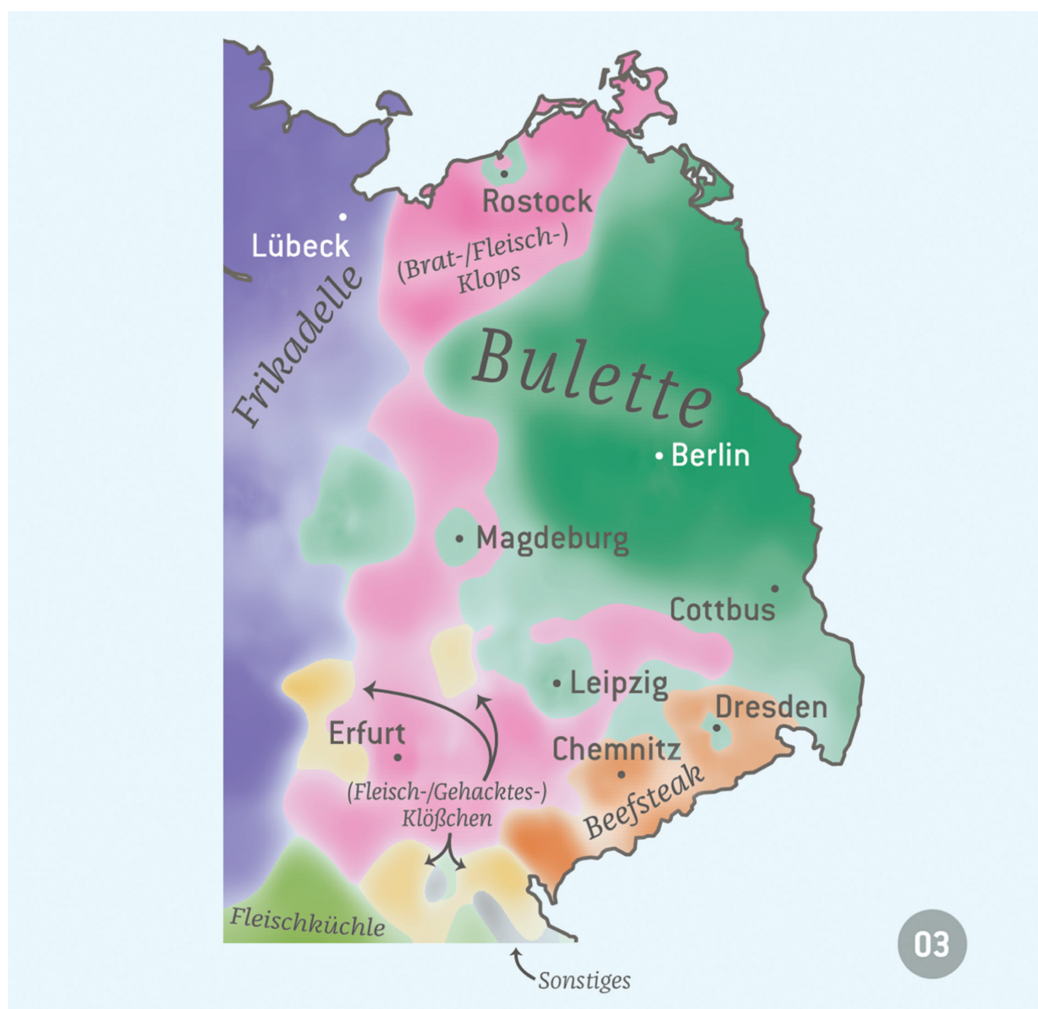


Figure 7: Map for the variants of “beef patty” for Northeastern Germany.



Figure 8: Augmented reality in Voice Äpp's *Dialektpanorama* functionality.

Finally, Jannedy et al. (2017, 2018) have explored initial audio data from *Deutschklang*. They tested whether audio data crowdsourced through smartphones enables reliable analysis of spectral contrasts between fricatives. They compared speech recorded in laboratories to that crowdsourced through the app and applied various acoustic measures (e.g., center of gravity). They found that data collected through the app is usable for analyses of fricatives, i.e., crowdsourced data is ecologically valid for these purposes. One (big) caveat is that the *Deutschklang* database was pruned manually to select speakers who had recorded themselves in quiet environments, meaning that a substantial portion of the *Deutschklang* recordings had already been discarded.

6 Future directions

As for the future of smartphone apps, tech pundits predict a number of trends continuing in the coming years, such as the shift toward augmented and virtual reality, artificial intelligence, wearables, beacons, chatbots, and the Internet of Things (IoT) (i.e., everything is connected to the internet – the fridge, home, radio, cars etc.) (cf. Entrepreneur 2019). In terms of augmented reality, *Voice Äpp* already makes use of this technology to visualize regional variation: in *Dialektpanorama* – one of the functionalities developed to increase user engagement with *Voice Äpp* – participants lift up the phone to display regional variants around them. Users can adapt the distance they want to show with a lever (Figure 8).

In the future, augmented and virtual reality may be used to show isoglosses in real geographical settings, displayed on the screen while hiking through different dialect regions, for example. Artificial intelligence-enabled chatbots implemented in smartphones may be useful for language learning and/or revitalizing languages. The *Sägemol* app by Slowsoft (Holm 2019), for example, translates input text from German and outputs that via text-to-speech to Zurich or Grisons Swiss German. How smartphone apps for linguistic purposes will evolve in terms of wearables or the IoT is difficult to predict. Perhaps there will be a move away from native apps to progressive web-based applications – i.e., web applications that have a native app feel and functionality but run in a smartphone app's browser (cf. Saeed 2018). Leemann and Yoo (2019) are presently trying out such a progressive web-app, called *emotionresponse*, which supports recording capability in a web browser and prompts users to record intuitive vocal responses to different emotions (shock, anger, happiness etc.) in typologically different languages.

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